

3. Requirements

3.1 Functional Requirements

The following table defines the functional requirements of the Police Management System (PMS), derived from the user scenarios, use cases, and user stories defined in this specification. Each requirement is assigned a unique identifier, a priority level, and describes what the system shall do.

Priority Scale: 1 = Must Have (Highest) | 2 = Should Have | 3 = Nice to Have (Lowest)

Req #	Requirement	Comments	Priority
FR_01	The system shall provide role-specific dashboards for each user type: Administrator, Police Officer, Detective, Forensic Expert, Traffic Officer, and Citizen.	Every role has its own dashboard displaying only relevant features and data.	1
FR_02	User passwords must be stored securely using a cryptographic hashing algorithm.	Passwords are saved as encrypted hashes rather than plain text.	1
FR_03	The system shall authenticate staff users using Admin-assigned credentials; Citizens may self-register.	Upon successful login, the system redirects the user to their role-specific dashboard.	1
FR_04	The system shall lock or flag an account after a defined number of consecutive failed login attempts.	Failed login attempts must be logged; repeated failures trigger protective account lockout.	1
FR_05	The system shall restrict access to all features and data based on the user's assigned role.	Role-based access control enforced at system level; users cannot perform actions outside their permitted scope.	1
FR_06	Administrator shall be able to create new user accounts for Police Officers, Detectives, Forensic Experts, and Traffic Officers.	Admin provides the new user's full name, username, password, role, and department during account creation.	1
FR_07	Administrator shall be able to update, deactivate, or reactivate existing user accounts.	Status changes take effect immediately; affected users are notified with a documented reason.	1
FR_08	Administrator shall be able to assign and modify user roles within the system.	Role changes take effect immediately and update the user's access permissions accordingly.	1
FR_09	Administrator shall be able to add and manage police departments within the system.	Departments can be created, edited, and deleted; users are organized by department.	1

Req #	Requirement	Comments	Priority
FR_10	Administrator shall be able to assign Police Officers to specific cases.	Admin selects a case and an available Officer; the Officer is notified of the assignment immediately.	1
FR_11	Administrator shall have access to a real-time analytics dashboard displaying platform-wide metrics.	Dashboard shows total cases per status, registered users per role, active departments, and recent incidents.	2
FR_12	Administrator shall be able to manage access control and permissions across all roles.	Admin can view and adjust what each role is permitted to see and do within the system.	1
FR_13	Police Officers shall be able to create new incident reports for reported crimes.	The report form requires: location, crime type, detailed description, date, and time of the incident.	1
FR_14	Police Officers shall be able to register and document arrest information linked to an existing case.	Arrest records include arrestee details, date/time of arrest, charges, and the arresting Officer's identity.	1
FR_15	Police Officers shall be able to manage patrol scheduling and assignments.	Officers can view and update their patrol schedule, including routes and time slots.	2
FR_16	Police Officers shall be able to register and manage complaints submitted by citizens or in person.	Complaints are linked to a citizen or case record and assigned for follow-up.	1
FR_17	Police Officers shall be able to manage records of persons of interest or suspects.	Officer can add, view, and update suspect profiles linked to active cases.	1
FR_18	Police Officers shall be able to write and submit investigation reports.	Reports are linked to a specific case and include narrative, findings, and actions taken.	1
FR_19	Detectives shall be able to open and manage criminal cases.	Detectives can initiate case records, assign investigators, and track case progress.	1
FR_20	Detectives shall be able to upload and link evidence files to an existing case.	Evidence is stored securely and associated with the correct case.	1
FR_21	Detectives shall be able to manage witness records and testimonies.	Witness entries include identity details, testimony content, and confidentiality status.	1
FR_22	Detectives shall be able to assign suspects to an existing case by entering the suspect's details.	Required fields: full name, age, physical description, and any related evidence references.	1

Req #	Requirement	Comments	Priority
FR_23	Detectives shall be able to track and update investigation status.	Status transitions (Open, Under Investigation, Closed) are logged with timestamp and actor identity.	1
FR_24	Detectives shall be able to write and submit investigation reports.	Reports include case summary, evidence analysis, suspect assessment, and investigative conclusions.	1
FR_25	Forensic Experts shall be able to perform and document DNA analysis on submitted samples.	Analysis results are linked to the relevant case and include methodology and findings.	1
FR_26	Forensic Experts shall be able to perform and document fingerprint analysis.	Fingerprints are analyzed and compared; results are stored linked to the case.	1
FR_27	Forensic Experts shall be able to perform and document weapon analysis.	Includes ballistics, tool marks, and trace evidence linked to submitted weapons.	1
FR_28	Forensic Experts shall be able to analyze and annotate photos and videos submitted as evidence.	Analysis includes metadata extraction, image enhancement notes, and findings.	1
FR_29	Forensic Experts shall be able to manage laboratory evidence (intake, processing, chain of custody).	System tracks evidence from submission through analysis and final storage.	1
FR_30	Forensic Experts shall be able to write and submit forensic reports.	Reports are formally linked to a case and include all analysis findings and conclusions.	1
FR_31	Traffic Officers shall be able to register traffic accidents.	Accident records include location, vehicles involved, injuries, witness information, and description.	1
FR_32	Traffic Officers shall be able to issue and register fines for traffic violations.	Fine records include violation type, amount, date, citizen/vehicle identification, and issuing officer.	1
FR_33	Traffic Officers shall be able to manage road violations and infractions.	Officers can view, update, and close violation records.	1
FR_34	Traffic Officers shall be able to perform and document vehicle control checks.	Checks include registration, insurance, and inspection status linked to a vehicle record.	2
FR_35	Traffic Officers shall be able to generate traffic reports and statistics.	Reports cover violation trends, accident frequency, and fine collection summaries.	2
FR_36	Traffic Officers shall be able to manage driving license records and validations.	Officers can search, view, and flag license records; suspension notes are recorded.	2

Req #	Requirement	Comments	Priority
FR_37	Citizens shall be able to report incidents through the public citizen portal.	Incident report form includes description, location, date/time, and optional photo attachment.	1
FR_38	Citizens shall be able to submit complaints regarding police conduct or services.	Complaint form includes category, description, and the relevant officer or department.	1
FR_39	Citizens shall be able to report missing persons.	Report includes personal details, last known location, date missing, and contact information.	1
FR_40	Citizens shall be able to check the status of their reported cases.	Citizens can view the current status of any case they have submitted.	1
FR_41	Citizens shall be able to view any traffic fines issued to them by Traffic Officers.	Citizens see fine details including violation type, amount, date, and payment status.	1
FR_42	Citizens shall be able to view system notifications and updates relevant to their reports.	Notifications are delivered when their report status changes or a response is added.	2
FR_43	The system shall enforce real-time form validation across all submission forms.	Validation executes client-side in real time without requiring a full page reload.	1
FR_44	All notifications triggered by system events shall be delivered to relevant parties within 1 minute.	Applies to: case assignments, status changes, suspect additions, and account changes.	1
FR_45	The system shall log all key actions with timestamps for audit purposes.	Logs include: login attempts, account changes, case creation, status updates, evidence uploads.	1
FR_46	The web application shall be responsive and accessible from any modern device.	Interface adapts to desktop, tablet, and mobile screen sizes without loss of functionality.	2

3.2 Non-Functional Requirements

3.2.1 Product Requirements

3.2.1.1 User Interface Requirements

The user interface for the Police Management System shall be compatible with all modern web browsers and accessible from both desktop and mobile devices. The interface is divided into three role-specific views, each tailored to the responsibilities of the assigned user.

Admin Interface

- A main dashboard displaying real-time platform metrics: total cases per status (Open, Under Investigation, Closed), registered users per role, active departments, and recent incidents
- A user management panel for creating, editing, deactivating, and reactivating Police Officer and Detective accounts, with role and department assignment controls
- A department management panel for adding, editing, and removing police departments, and viewing officers organized by unit
- A case overview panel displaying all cases in the system, filterable by status, assigned officer, department, and date range
- A case assignment panel allowing the Admin to assign available Police Officers to specific open or ongoing cases
- An analytics and statistics section displaying crime trends, case resolution rates, and department performance metrics
- A notifications panel displaying all system-generated alerts in reverse chronological order

Police Officer Interface

- A personal dashboard displaying all cases currently assigned to the Officer, with their statuses, dates, and quick-access links
- An incident report creation form with required fields: location of the incident, crime type, detailed description, date, and time
- A case management panel allowing the Officer to view full case details, update case information, update case status, and add notes
- An evidence upload panel allowing the Officer to attach and manage files (documents, images, videos) linked to a specific case
- An arrest record form allowing the Officer to document arrest details, charges, date/time of arrest, and link the record to the relevant case
- A notifications panel displaying all system-generated alerts relevant to the Officer in reverse chronological order

Detective Interface

- A personal dashboard displaying all cases assigned to the Detective, with full case details and investigation history
- A suspect management panel allowing the Detective to add, view, and manage suspect records linked to an existing case
- A suspect entry form with required fields: full name, age, physical description, and related evidence references
- An evidence panel allowing the Detective to upload and manage evidence files linked to assigned cases
- A case review section where the Detective can view all incident reports, evidence, suspect entries, and arrest records associated with their assigned cases
- A notifications panel displaying all system-generated alerts relevant to the Detective in reverse chronological order

Forensic Expert Interface:

- A dashboard displaying pending and completed forensic analysis requests
- Analysis panels for DNA, fingerprints, weapons, and photo/video evidence

- A laboratory evidence management panel tracking chain of custody
- A forensic reports section for writing and submitting formal forensic reports
- A notifications panel for case-related alerts and new evidence submissions

Traffic Officer Interface:

- A dashboard displaying recent accidents, violations, and pending actions
- An accident registration form (location, vehicles, injuries, description)
- A fine management panel for issuing and tracking traffic fines
- A violations management section for recording and managing road infractions
- A vehicle control section for documenting control checks
- A license management section for validating and flagging driving licenses
- A reports section for generating traffic statistics and summaries

Citizen Interface:

- A public-facing portal accessible after self-registration
- An incident reporting form (description, location, date/time, optional photo)
- A complaints submission form (category, description, relevant officer/departments)
- A missing person report form (personal details, last known location, date missing)
- A case status tracker for monitoring submitted reports
- A fines viewer for checking traffic fines issued to the citizen
- A notifications section for updates on submitted reports

3.2.1.2 Learnability

The Police Management System shall be designed so that users across all three roles can operate the system effectively with minimal prior training. Key learnability features include:

- **Intuitive Role-Based Interfaces:** Each role's dashboard surfaces only the features and data relevant to that role, reducing cognitive load and preventing confusion from irrelevant options
- **Consistent Navigation Patterns:** Common actions such as submitting reports, updating case statuses, and managing suspects follow standardized workflows and button placement across all role interfaces
- **Contextual Guidance:** Key actions such as creating an incident report, adding a suspect, or uploading evidence shall include inline instructions or tooltips explaining required fields and constraints
- **Clear Status Indicators:** All cases shall display their current status (Open, Under Investigation, Closed) prominently and consistently, using standardized labels and color coding
- **Minimal Training Dependency:** The interface shall rely on familiar web design patterns — form submissions, confirmation dialogs, notification badges — to reduce the need for formal onboarding documentation

3.2.1.3 Accessibility

The Police Management System shall adhere to accessibility standards to ensure usability for all users:

- **Standards Compliance:** The web interface shall follow Web Content Accessibility Guidelines (WCAG) 2.1, including sufficient color contrast ratios, full keyboard navigability, and compatibility with screen reader technologies
- **Responsive Design:** The platform shall be fully functional across desktops, tablets, and mobile devices, adapting layouts appropriately to varying screen sizes without loss of functionality
- **Text Alternatives:** All images, icons, and non-text UI elements shall include descriptive alt text for screen reader compatibility
- **Adjustable Text:** The interface shall remain functional and readable when users adjust browser zoom levels or font size settings
- **Accessible Forms:** All report creation, suspect entry, and action forms shall include properly labeled fields, clear error messages, and logical tab order to support users relying on assistive technologies

3.2.1.4 Efficiency

The platform shall be optimized to support fast task completion across all user roles:

- **Page Load Performance:** All pages and dashboards shall load within 3 seconds under normal network conditions; login and messaging interfaces shall load within 2 seconds
- **Case Search Performance:** Case search and filter results shall update within 2 seconds of the user applying or modifying a filter, without requiring a full page reload
- **Notification Delivery:** All system-triggered notifications (case assignments, status changes, account changes) shall be delivered to relevant parties within 1 minute of the triggering event
- **Inline Validation:** All forms shall provide real-time client-side validation, highlighting errors immediately without requiring form submission, reducing correction cycles
- **Optimized Workflows:** Frequently performed actions — such as updating a case status, submitting an incident report, and adding a suspect — shall be accessible within a minimal number of navigation steps from the user's dashboard

3.2.1.5 Dependability and Memorability

The PMS shall ensure that users can return after periods of inactivity and quickly re-establish proficiency:

- **Consistent Layout:** Core interface elements such as navigation menus, dashboards, notification panels, and action buttons shall maintain consistent placement and behavior across all sessions and screen sizes
- **Standardized Visual Language:** Icons, status labels, and color indicators shall be used consistently throughout the platform — for example, case status labels always use the same terminology and color coding
- **Persistent Notification History:** All system-generated notifications shall be retained and accessible from the user's notification panel, allowing users to review past events and case milestones after returning to the platform
- **Case Context Preservation:** Users returning mid-investigation shall find all relevant case history, evidence records, suspect entries, and audit logs intact and accessible from their dashboard without requiring re-navigation

- **Error Recovery Guidance:** When users make errors — such as submitting an incomplete report or attempting an action outside their permissions — the system shall display clear, descriptive messages guiding the user toward a valid next step

3.2.1.6 Errors

The PMS shall minimize user errors and provide clear, structured recovery mechanisms:

- **Preventive Validation:** Input validation shall block logically invalid entries before submission — for example, preventing a case status update to 'Closed' without the required authorization, or a suspect assignment to a non-existent case
- **Descriptive Error Messages:** All error messages shall clearly explain the issue and suggest a corrective action, avoiding technical jargon — for example, 'This case does not exist. Please verify the Case ID before assigning a suspect.'
- **Confirmation Dialogs:** All irreversible or high-impact actions — including account deactivation, case closure, and suspect deletion — shall require explicit user confirmation through a dialog prompt before being processed
- **Permission Boundary Enforcement:** When a user attempts an action outside their role's permitted scope, the system shall block the action at the system level and display a clear, non-technical message informing the user that the action is not available to their role
- **Error Logging:** All system errors and failed user actions shall be logged with timestamps for administrative review and troubleshooting, with no sensitive user data exposed in error outputs
- **Graceful Degradation:** If a non-critical feature such as file upload experiences a temporary failure, the system shall retain all other form data entered by the user, notify them of the specific failure, and prompt a retry without requiring the user to restart the entire form

3.2.1.7 Satisfaction

The Police Management System is designed to provide a clear, professional, and efficient experience for all three user roles — from Police Officers documenting frontline incidents, to Detectives managing complex investigations, to Forensic Experts conducting scientific analysis of evidence, to Traffic Officers handling road safety and violations, to Citizens interacting with the system for reporting and status tracking, and to Administrators overseeing the entire system.

The role-specific dashboards ensure that each user sees only what is relevant to them, reducing friction and making routine tasks — such as filing an incident report, updating a case status, or assigning a suspect — straightforward and immediate. The real-time notification system keeps all parties informed at every stage of an investigation without requiring manual follow-up, reinforcing accountability and operational transparency.

Because the PMS is fully web-based and responsive, users can access their dashboards and take action from any device, ensuring that time-sensitive steps — such as updating a case status immediately after an arrest or submitting an incident report from the field — are never blocked by device or location constraints.

3.2.1.8 Capacity

The Police Management System shall be developed to support the concurrent operation of all three user roles without performance degradation. All users interact with a shared centralized database, and the system shall process concurrent requests reliably, ensuring that case status changes, new report submissions, and account updates are reflected in real time for all authorized parties.

The platform is web-based and does not integrate with external national law enforcement databases or third-party services in its current scope, which limits external dependency points and reduces the risk of third-party performance bottlenecks. All data — cases, incident reports, evidence, suspects, arrest records, and audit logs — shall be stored in a centralized database with regular automated backups to prevent data loss in the event of system failure.

3.2.2 Organizational Requirements

3.2.2.1 Availability

The Police Management System shall be available at all times to support ongoing criminal investigations, which may require user action outside of standard business hours. Specific availability requirements include:

- The platform shall target a minimum uptime of 99% during active deployment and operational phases
- Scheduled maintenance windows shall be communicated to users in advance where possible and minimized in duration to reduce operational disruption
- Regular automated database backups shall be performed to ensure data integrity and enable full recovery in the event of system failure

3.2.2.2 Latency

Response times for all operations shall meet the following thresholds under normal network conditions:

- Page and dashboard loads: within 3 seconds
- Login interface load: within 2 seconds
- Form submissions (incident report creation, suspect assignment, evidence upload): within 5 seconds
- Case search and filter result updates: within 2 seconds of filter application
- Notification delivery: within 1 minute of the triggering event
- Account status changes and case assignments by Admin: reflected immediately upon confirmation

3.2.2.3 Monitoring

The platform shall support operational monitoring through the following mechanisms:

- The Admin analytics dashboard shall display real-time platform metrics including total cases per status, registered users per role, active departments, and recent incident activity, updated with no more than a 1-minute delay
- All key system events — login attempts, account changes, case creation, status updates, evidence uploads, suspect assignments, arrest records, and case closures — shall be logged with timestamps and accessible for administrative review
- System errors and failed operations shall be logged automatically and retained for troubleshooting and audit purposes

3.2.2.4 Maintenance

- Regular software updates shall be scheduled to address security vulnerabilities, performance improvements, and bug fixes

- System audit logs shall be maintained and retained for the duration of active investigations and for a defined period thereafter to support legal accountability and compliance review
- All case records, incident reports, evidence files, suspect entries, and arrest records shall be retained in the system even after a case is closed, to support future dispute, audit, or legal proceedings

3.2.2.5 Operations

The Police Management System is designed to be operable by users across a range of technical proficiency levels, from Administrators managing the platform daily to Police Officers who may primarily interact with it for report submission and status updates. The interface relies on familiar web interaction patterns and requires no software installation beyond a standard web browser. All core workflows are self-contained within the platform, and no external tools or integrations are required for any user to complete their role-specific tasks.

3.2.2.6 Standards Compliance

The Police Management System shall comply with the following standards and regulations:

- **GDPR (General Data Protection Regulation):** All personal data collected during account creation and platform use shall be processed and stored in compliance with GDPR requirements, including purpose limitation, data minimization, and user rights regarding their personal data
- **WCAG 2.1:** The user interface shall conform to Web Content Accessibility Guidelines 2.1 to ensure accessibility for users with disabilities
- **Password Security Standards:** All user passwords shall be stored as cryptographic hashes using a modern algorithm (e.g., bcrypt); plain-text password storage is strictly prohibited
- **Audit Trail Requirements:** All key platform actions shall be logged with timestamps and actor identifiers to support accountability and legal compliance in the event of disputes, regulatory review, or court proceedings

3.2.2.7 Portability

- The PMS shall be delivered as a web-based application, ensuring consistent operation across all major operating systems including Windows, macOS, Linux, iOS, and Android
- Users shall require only a device with a modern web browser and a stable internet connection to access the full platform
- The platform shall be fully functional across all commonly used browsers including Chrome, Firefox, Safari, and Edge
- No client-side software installation shall be required for any user role to access or operate the platform

3.2.3 External Requirements

3.2.3.1 Protection

To protect the Police Management System and its users from unauthorized access, data breaches, and misuse, the following security measures shall be implemented:

- All user passwords shall be stored using a cryptographic hashing algorithm; plain-text storage is prohibited under any circumstance
- All data transmitted between the client and server shall be encrypted using HTTPS with TLS to prevent interception

- All user inputs across all forms shall be validated both client-side and server-side to prevent injection attacks, cross-site scripting, and other input-based vulnerabilities
- Role-based access control shall be enforced at the system level, ensuring users can only access data and features within their permitted scope regardless of direct URL access or API manipulation
- All irreversible actions shall require explicit user confirmation to prevent accidental data loss or unauthorized modifications
- Each user shall only be able to access their own assigned cases, reports, and records; no cross-user data visibility exists outside of the defined role permissions
- Session tokens shall expire after a defined period of inactivity, requiring re-authentication to resume access
- The system shall lock or flag an account after a defined number of consecutive failed login attempts to protect against brute-force attacks

3.2.3.2 Authorization and Authentication

- User authentication shall be performed using username and password credentials assigned by the Administrator, validated server-side on each login attempt
- The system shall use JSON Web Tokens (JWT) or an equivalent secure session management mechanism to maintain authenticated sessions
- Authorization shall be enforced based on user role at the system level, ensuring that each user can access only the features, data, and actions defined for their role
- All failed login attempts shall be logged with timestamps and the system shall apply protective lockout measures after a defined threshold of consecutive failures
- Admin accounts shall have the highest level of access; no other role can access user management, department management, or system-wide analytics

3.2.3.3 Legislative Requirements

The Police Management System operates in a law enforcement domain involving sensitive personal data, criminal records, suspect information, and legal evidence. The following legislative requirements apply:

- **Data Protection:** The platform shall comply with applicable personal data protection legislation, including GDPR, ensuring that all user and case data is collected only for defined purposes, stored securely, and not accessible by unauthorized parties
- **Evidence and Record Integrity:** All case records, incident reports, evidence files, and arrest records shall be immutable once formally submitted and locked, as they may carry legal implications in criminal proceedings
- **Audit Accountability:** All key platform actions shall be logged and retained to support accountability requirements in the event of disputes, regulatory review, or legal proceedings
- **Right to Data Access:** Users shall be able to access their own account information and assigned case data through the platform in compliance with data subject access rights defined under applicable legislation

3.3 Domain Requirements

The Police Management System operates within the law enforcement management domain, where the primary goal is to digitize and structure the lifecycle of criminal investigations — from initial incident reporting through case management, evidence tracking, suspect assignment, and final case closure.

The domain imposes specific constraints on the platform's behavior. Criminal investigations are high-stakes processes involving legal obligations, sensitive personal data, and public safety considerations. As a result, the platform prioritizes data integrity, auditability, and strict role separation above convenience features. Arrest records and incident reports are permanently retained once submitted, as they may carry legal implications in criminal proceedings. Case histories and all status transitions are fully logged for the same reason.

The platform does not handle legal contract generation, court case management, or external law enforcement database integration in its current scope, recognizing that these aspects are governed by regulatory and legal frameworks that fall outside the boundaries of a web-based management platform. All formal legal proceedings following case closure are assumed to occur through appropriate external channels.

Access control and data privacy are of particular importance in this domain, as the platform handles sensitive information including suspect details, criminal evidence, and officer activity records. Each user role has strictly defined visibility boundaries — Police Officers can only view cases assigned to them, Detectives cannot access cases outside their assignment, and the Admin has system-wide oversight but cannot perform investigative actions on behalf of Officers or Detectives — reflecting both privacy requirements and the professional accountability structures of law enforcement.